

Digital Signal Intelligence Revolutionary Technical Pricing Model

Using enterprise digital footprint analysis to transform underwriting economics and risk selection

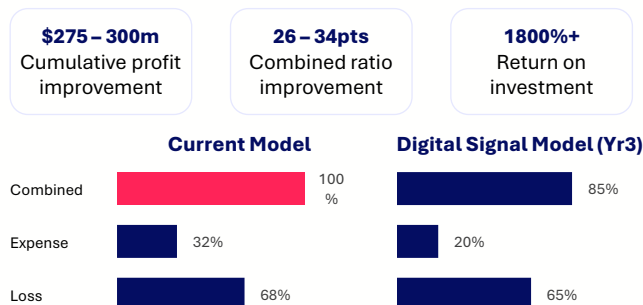
THE CORE THESIS

A company's digital presence—how they maintain their website, who they're connected to online, and what they disclose—serves as a powerful proxy for operational maturity, governance quality, and risk management capability. By systematically scoring these "digital signals," we can automate 75-85% of underwriting decisions while simultaneously improving risk selection, enabling zero-touch pricing for most submissions.



This isn't about adding automation to existing models, it's about replacing manual underwriting with an entirely new data pillar that works globally, updates continuously, and costs 90% less than traditional methods.

5-YEAR FINANCIAL IMPACT



HOW DIGITAL SIGNALS WORK

We extract and quantify four signal categories from public web data:

- Infrastructure signals**
SSL certificates, security posture, technology stack (predicts cyber exposure)
- Content signals**
Governance transparency, disclosure depth, policy visibility (predicts D&O risk)
- Network signals**
Link quality, vendor ecosystem, supply chain connections (predicts credit/political risk)
- Behavioural signals**
Maintenance discipline, incident response speed, operational consistency (predicts all lines)

D&O
12 – 18pt LR improvement

Cyber
15 – 25pt LR improvement

E&O
10 – 15pt LR improvement

Credit
Early warning system

Marine
Risk segmentation

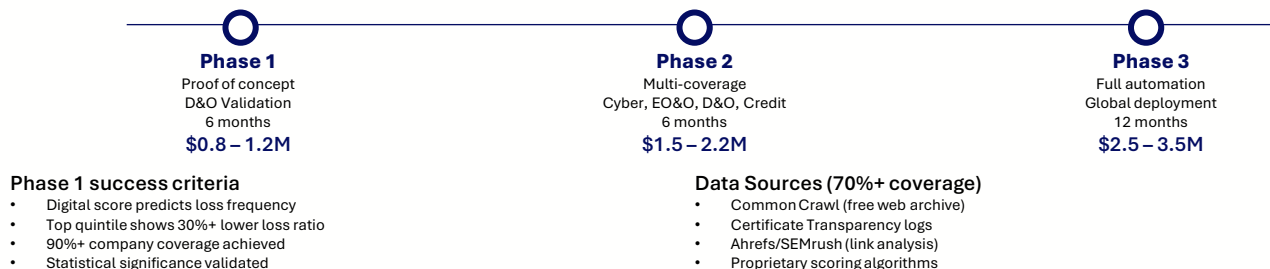
OPERATIONAL TRANSFORMATION: FROM 2 WEEKS TO 4 HOURS

	Current state	Digital signal model
Timeline	2 - 14 days	45 mins – 4 hours
Touch-points	8 – 15 manual steps	0 – 2 (exceptions only)
Cost per quote	\$650	\$70
Hit-ratio	35%	45%
Note	Manual data gathering, surveys, underwriter review, multiple approval layers	Automated lookup, instant risk scoring, rule-based decision making, 75 – 85% straight-through



\$25.7M Annual Expense Savings
On 50,000 submissions (typical mid-carrier market volume)

PHASED IMPLEMENTATION: RE_RISKED APPROACH



WHY THIS MATTERS NOW

- First-Mover Window**
18-36 months before competitors replicate; regulatory approvals create barriers
- Market Expectations**
Instant quotes are table stakes in 2025; 2-week turnarounds lose business
- Talent Magnet**
Talent won't join "insurance as usual" companies; need cutting-edge projects
- Margin Compression**
Without operational gains, target ROE becomes unachievable in competitive markets

DEFENSIBLE COMPETITIVE ADVANTAGES

- Data Moat**
Proprietary historical validation and signal calibration (3+ years to replicate)
- Network Effects**
More scored companies = more accurate relative positioning
- Regulatory Lead**
Rate filings and actuarial documentation create 2-3-year barriers
- Learning Loop**
Claims feedback improves model faster than competitors starting from scratch

BOTTOM LINE FOR THE BOARD

Approve phase 1 – proof of concept
\$800K - \$1.2M

6-month validation of D&O coverage – definitive evidence of predictive power, decision point before commitment

Success outcome
Fast-track phase 2
Immediate competitive advantage

Neutral outcome
Refine and retry
Limited downside, learning value

Failure outcome
Stop investment
\$1M loss, clear market signal

Success Probability

70-80% that digital signals show statistically significant predictive power for 2-3+ coverage lines based on academic research on organisational digital maturity and preliminary data exploration

Risk-Adjusted Expected Value

\$175-280M (70% probability × \$250-400M potential 5-year value)

Digital Signal Intelligence

The PageRank precedent: why this is proven, not experimental

How network analysis revolutionized search in 1998, and why the same methodology will transform insurance pricing in 2025

1998: GOOGLE'S REVOLUTIONARY INSIGHT

The problem	Search engines ranked pages by counting keywords. Result: Spam, gaming, and irrelevant results dominated search.
Googles solution	PageRank: don't analyse what pages say about themselves; analyse who links to them and the quality of those links. A page linked by trusted sources must itself be trustworthy.
The impact	Google grew from startup to \$2 trillion company by proving that network position and relationship quality predict value better than self-reported information.

THE DIRECT ANALOGY TO INSURANCE PRICING

	PAGERANK (1998) RANKING WEB PAGES	SIGNAL SIGNAL INTELLIGENCE (2025) RANKING COMPANIES FOR RISK
Challenge	Which websites are trustworthy and authoritative?	Which companies are well-managed and lower risk?
Solution	Analyse link networks, connection quality, and digital infrastructure signals	Analyse same link networks, connection quality, and digital infrastructure signals
Result	92% global search market share	26-34pt combined ratio improvement



Google proved that who you're connected to, how you maintain your digital presence, and your position in the network reveals more truth than what you claim about yourself. We're applying this exact same proven methodology to insurance underwriting.

HOW PAGERANK ACTUALLY WORKS

- Link Quality Over Quantity**
One link from MIT matters more than 100 links from spam sites → One partnership with a Fortune 500 company signals more than 100 generic vendor relationships
- Network Position Reveals Standing**
Pages at the center of quality networks rank higher → Companies connected to well-managed peers likely well-managed themselves
- Infrastructure Signals Credibility**
HTTPS, fast load times, clean code indicate professionalism → Same infrastructure signals predict operational discipline
- Behaviour Over Time Matters**
Consistent maintenance and growth patterns indicate quality → Historical digital discipline predicts future risk management
- Difficult to Game**
Can't buy your way to quality links without actual value → Can't fake digital maturity without actual investment in operations

WHY THIS WORKS NOW (NOT IN 1998)

The web has fundamentally restructured to allow this analysis:

- Universal Business Presence**
85% of mid-market+ companies have meaningful web footprint (vs. 15% in 1998)
- Standardised Signals**
SSL certificates, security headers, technology stacks are now universal and machine-readable
- Rich Network Data**
Two decades of link-building created analysable relationship graphs spanning millions of companies
- Archive Infrastructure**
Internet Archive, Common Crawl provide historical snapshots enabling temporal analysis
- API Economy**
Real-time access to ranking data, security scores, and network intelligence via commercial APIs

1.1B

Websites Globally
(Analysable)

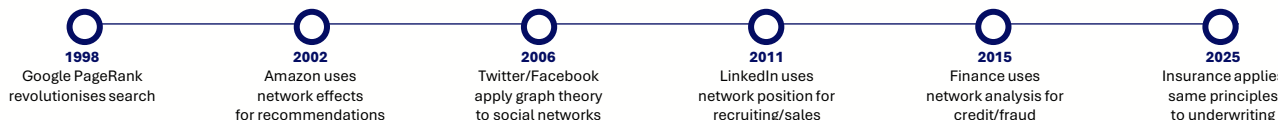
850M

SSL Certificates Issued
(Security Signal)

650B+

Indexed Pages (Network
Data)

HOW PAGERANK PRINCIPLES RESHAPED INDUSTRIES BEYOND SEARCH



PROVEN SUCCESS: INDUSTRIES THAT ALREADY USE NETWORK INTELLIGENCE

FICO & Credit Bureaus Application: Credit scoring now incorporates digital footprint, bill payment networks, and financial behaviour patterns mined from transaction graphs. Impact: Default prediction accuracy improved 35-40% over traditional credit scores alone.	Consumer Finance	Recorded Future Application: Predicts cyber attacks by analysing dark web networks, vulnerability disclosures, and company security posture signals. Impact: Customers detect threats 6-8 weeks earlier than traditional methods; now used by 1,800+ organizations.	Cybersecurity Intelligence
Palantir & Financial Crimes Application: Network analysis identifies money laundering by mapping transaction patterns, entity relationships, and connection quality. Impact: JPMorgan reduced false positives by 50% while catching 20% more actual fraud.	Banking/AML	Dun & Bradstreet Application: Business credit scores incorporate digital presence, payment networks, and supply chain relationships beyond traditional financials. Impact: 90% of Fortune 500 use network-enhanced credit scores for B2B decisions.	Commercial Credit/Risk
ZoomInfo & LinkedIn Application: Company scoring based on web presence, technology stack, hiring velocity, and network connections predicts sales opportunity quality. Impact: \$2B+ ARR business built entirely on digital signal intelligence.	B2B Sales Intelligence	Enigma Technologies Application: Small business lending uses web presence, online reviews, social signals, and network position to underwrite loans in minutes. Impact: Approval rates 2-3x higher than traditional methods with similar loss ratios; \$100M+ originated monthly.	FinTech/Lending

WHY THIS ISN'T SPECULATIVE

- 26 Years of Validation**
PageRank principles have been refined across search, social networks, finance, and fraud detection since 1998
- Academic Foundation**
Network science and graph theory are established fields with decades of peer-reviewed research
- \$100B+ Market Cap Built on It**
Google, LinkedIn, Palantir—all built massive businesses on network intelligence
- Infrastructure Exists**
Unlike 1998, the data layer, APIs, and analytical tools are all mature and accessible

OUR COMPETITIVE TIMING ADVANTAGE

- Proven Methodology**
Not inventing new science, instead applying 26-year-old, battle-tested approach
- Mature Infrastructure**
Data sources, analytical tools, and regulatory frameworks all exist today
- Insurance Lag**
While tech/finance adopted this 10-15 years ago, insurance hasn't—we can be first
- Regulatory Path**
Credit scoring, telematics, and other alternative data have established legal precedent
- Market Ready**
Instant quotes, digital underwriting, and data-driven pricing are the future

BOTTOM LINE FOR THE BOARD

This proposal isn't asking for a bet on unproven technology or experimental methodologies, instead the value is proven and has already:

- Built Google into a \$2 trillion company
- Powers fraud detection at every major bank
- Predicts cyber attacks weeks before they occur
- Drives B2B sales intelligence for 90% of Fortune 500
- Been used across multiple, disparate industries.